GM/ID SIMULATION OF NMOS & PMOS

DESIGN & LAYOUT OF CMOS BANDGAP VOLTAGE REFERENCE USING SKY 130 PDK

Submitted by
Group 4
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I. What is Gm/Id ?

The g_m/I_D method is a technique used to design analog circuits by choosing how efficiently a MOSFET should convert its input voltage change into an output current change.

g_m/I_D = **Transconductance Efficiency**

g_m = transconductance of the MOSFET, measured in siemens (S).
I_D = drain current flowing through the MOSFET, measured in amperes (A).
g_m/I_D = transconductance efficiency, measured in V^{-1}

g_m/I_D tells you how much **transconductance** you obtain for the amount of **drain current** you use.

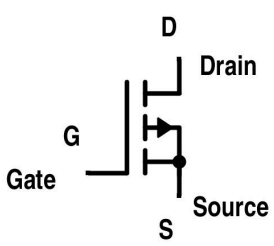
A higher g_m/I_D means you obtain more **transconductance per unit of drain current**. This is particularly useful when designing **low-power analog circuits.**

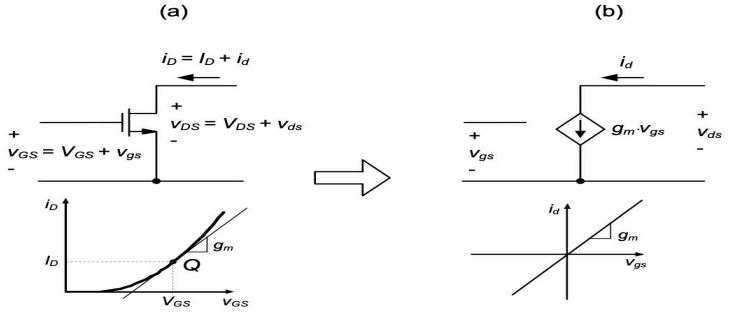
II. Gm Means?

Imagine that a MOSFET is a water-control valve.

I_D**: Drain curren**t : The current flowing from drain to source in an NMOS transistor. The gate voltage controls how much current flows.

g_m**: Transconductance :** It measures how much the drain current changes when the gate-source voltage changes slightly, with the other relevant bias conditions held constant.





In an analog circuit, it helps convert a small input-voltage signal into a current signal. That current can then produce a voltage signal through a load, helping create amplification.

III. Calculate g_m/I_D with an example

MOSFET A	MOSFET B
Drain current : 100 µA	Drain current : 100 µA
Transconductance : 2 mS	Transconductance : 2 mS
Efficiency : 20 V^{-1}	Efficiency : 10 V^{-1}

Both transistors use the same drain current, but MOSFET A provides twice the transconductance.

Efficiency : Gm/Id

Therefore, MOSFET A is more transconductance-efficient at these operating points.

Alternatively, if both transistors must provide $2mS$ of transconductance, a device operating at $20V^{-1}$ ideally needs only $100\mu A$, whereas one operating at $10V^{-1}$ needs $200\mu A$.

This is one of the main reasons analog designers use the g_m/I_D method.

IV. What is the actual purpose of the g_m/I_D method?

The main purpose is to choose the right operating point of a MOSFET before deciding its width and final circuit implementation.

In analog design, you often have specifications such as:

- Required gain
- Required power consumption
- Required speed or bandwidth
- Required drain current
- Required supply voltage

The g_m/I_D method helps you make design decisions that balance these requirements.

For example, suppose you are designing an operational amplifier (op-amp). You want high gain but low power consumption.

A larger g_m/I_D can help you obtain a given g_m using less current. However, this can come with lower speed and may require a larger transistor width.

The method lets you investigate these trade-offs using simulated transistor data rather than relying entirely on simplified textbook equations

V. What are the three operating regions?

1. *Weak inversion*

   $g_m/I_D > 20\ V^{-1}$

   - Very little current is needed for a given transconductance.
   - Suitable for low-power applications.
   - Usually offers lower speed and may require a larger width

2. *Moderate inversion*

   $g_m/I_D \sim 20\ V^{-1}$

   - Balances efficiency and speed.
   - Useful for many analog building blocks.
   - Often a practical starting region for amplifier design.

3. *Strong inversion*

   $g_m/I_D \sim 20\ V^{-1}$

   - Usually requires more current for the same g_m.
   - Can support higher-speed operation.
   - Often permits a smaller width for a given current, depending on the chosen operating point and process.

Important: High g_m/I_D is not always better. The correct region depends on your circuit's speed, power, gain and area requirements.

VI. What is V_{OV}, and why do we plot it against g_m/I_D?

V_{OV} means overdrive voltage. For an NMOS transistor, it is conventionally defined as

$$V_{OV} = V_{GS} - V_{TH}$$

V_{GS} : gate-to-source voltage.

V_{TH}: threshold voltage.

In a simplified long-channel model, g_m/I_D is approximately $2/V_{OV}$ in strong inversion. But real modern MOSFETs do not follow this relationship perfectly, especially outside strong inversion.

So it recommends using actual simulated transistor data instead of relying only on square-law calculations.

Tools Used Here

- Schematic : Xschem (*Open Source Tool - Alternative for Cadence Virtuoso*)
- PDK : SKY 130
- Simulation : NGSpice

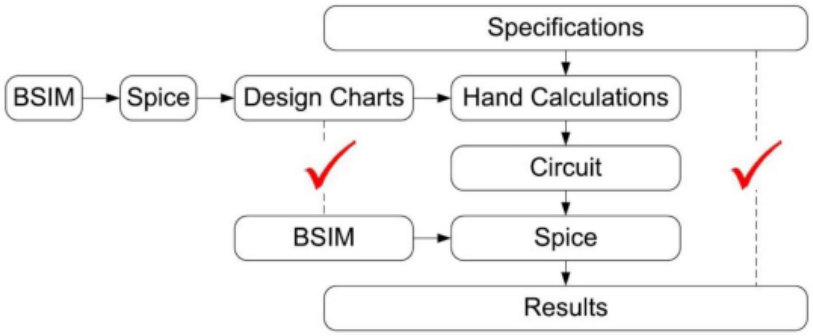
Plots in this Simulation

g_m/I_D **vs** V_{OV}	Shows transconductance efficiency
g_m/g_{ds} **vs** V_{OV}	Shows intrinsic gain capability
g_m/g_{ds} **vs** g_m/I_D	Helps select efficiency versus gain
f_T **vs** V_{OV}	Shows transistor speed versus overdrive voltage
f_T **vs** g_m/I_D	Shows the efficiency versus speed trade-off
I_D/W **vs** V_{OV}	Shows current density versus overdrive voltage
I_D/W **vs** g_m/I_D	Helps determine transistor width

W = Width is Fixed here
L = Length that we are varying here
Intrinsic Gain (g_m/g_{ds}) : The transistor's ability to provide voltage gain by itself.
Higher intrinsic gain generally means greater voltage-gain capability.
Transit Frequency (f_T) : A measure of how fast a transistor can operate.

$$f_T = g_m/2\pi C_{gg}$$

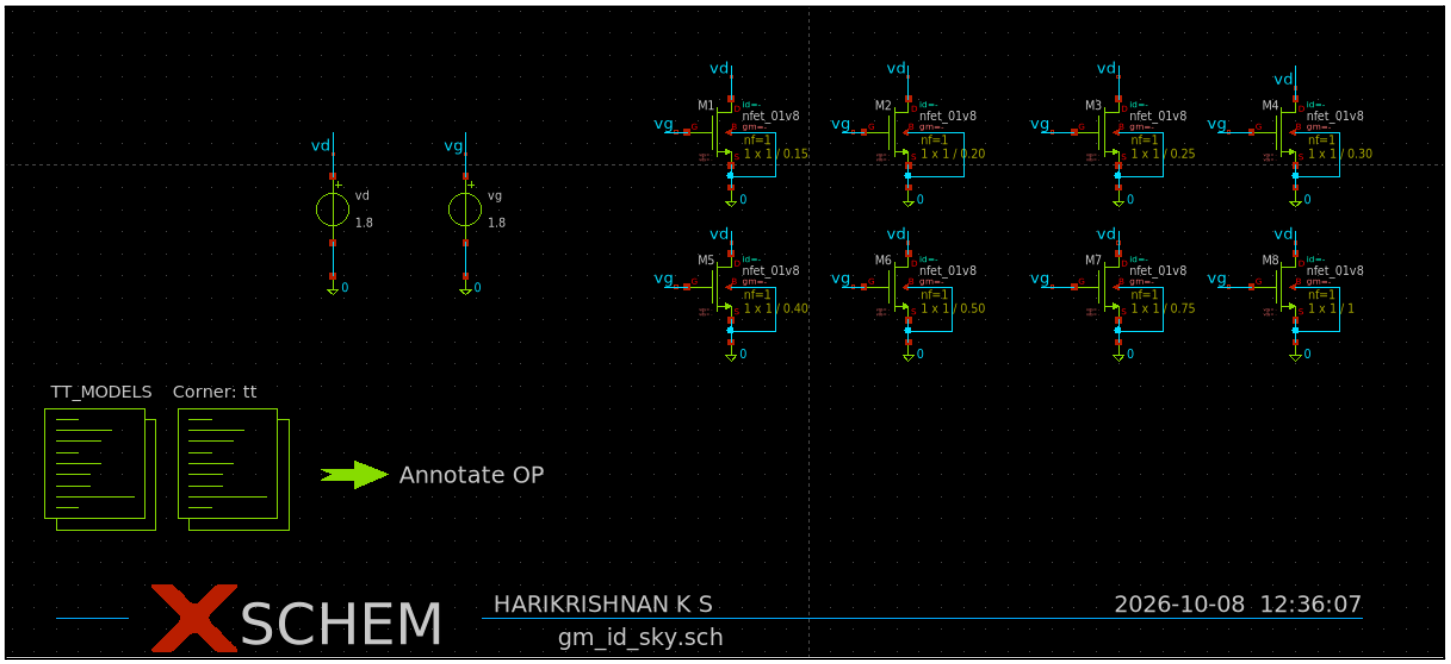
C_{gg} = Total gate capacitance of a MOSFET.
Current Density (I_D/W) : The drain current flowing per unit width of the transistor.



Credits : CuriousIC

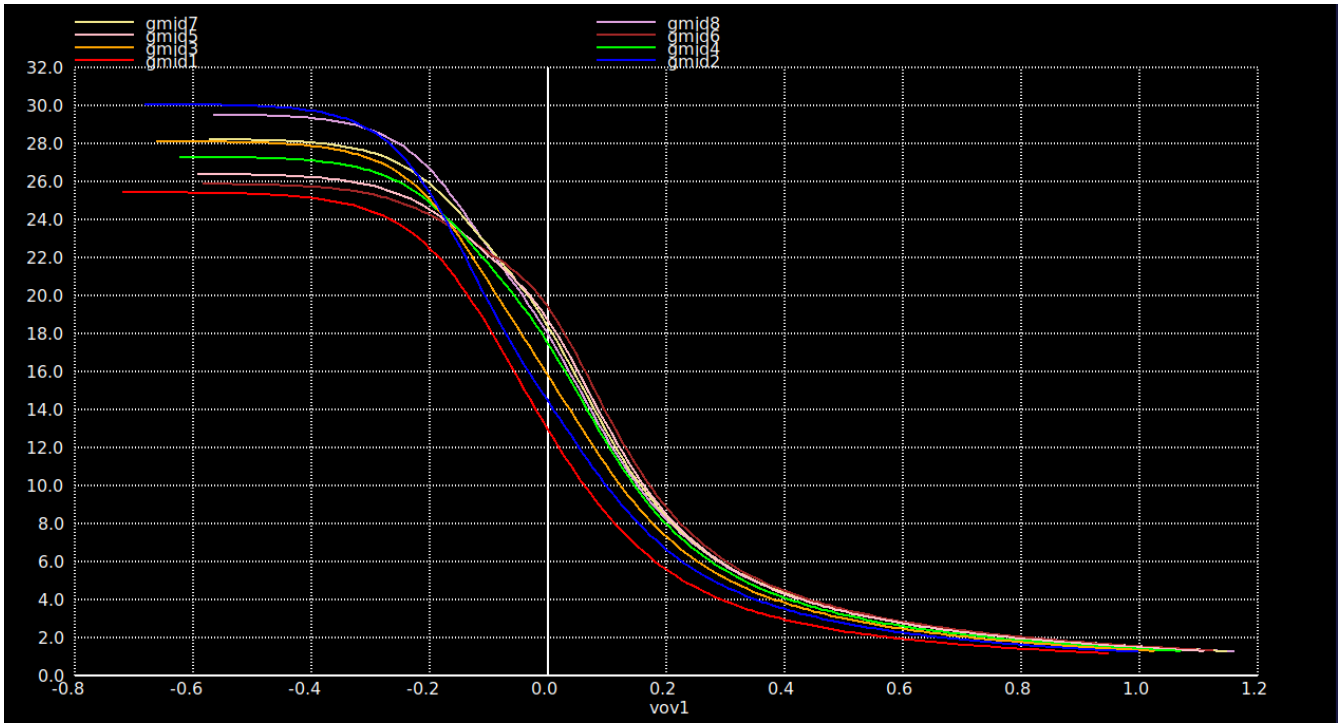
DESIGN CHARTS OR SIMULATIONS FOR NMOS

Length Varied here : 0.15u, 0.20u, 0.25u, 0.30u, 0.40u, 0.50u, 0.75u, 1u
Width is fixed here
Supply Voltage : 1.8V
Used FET from sky 130 PDK : nfet_01v8

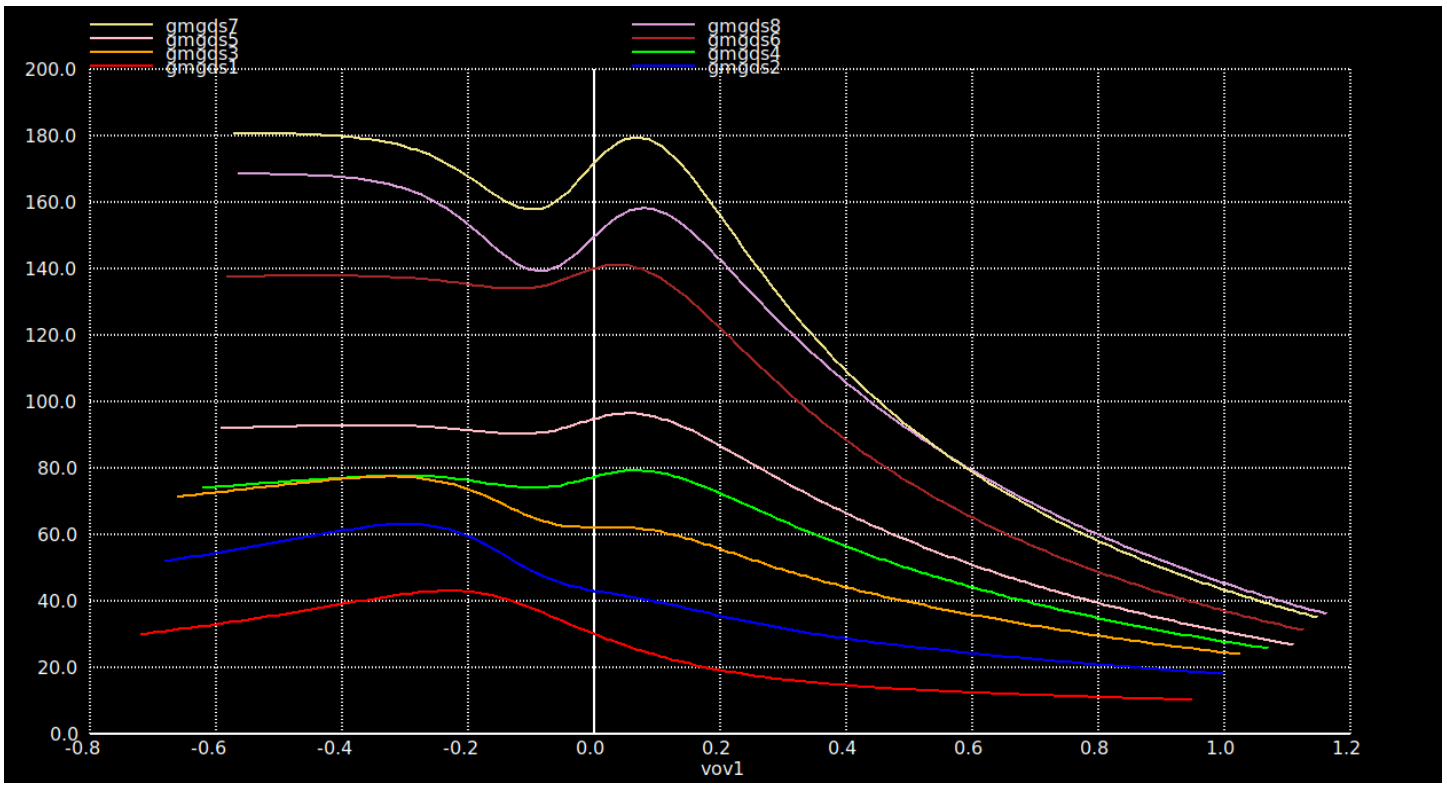


Schematic for Nmos Design charts

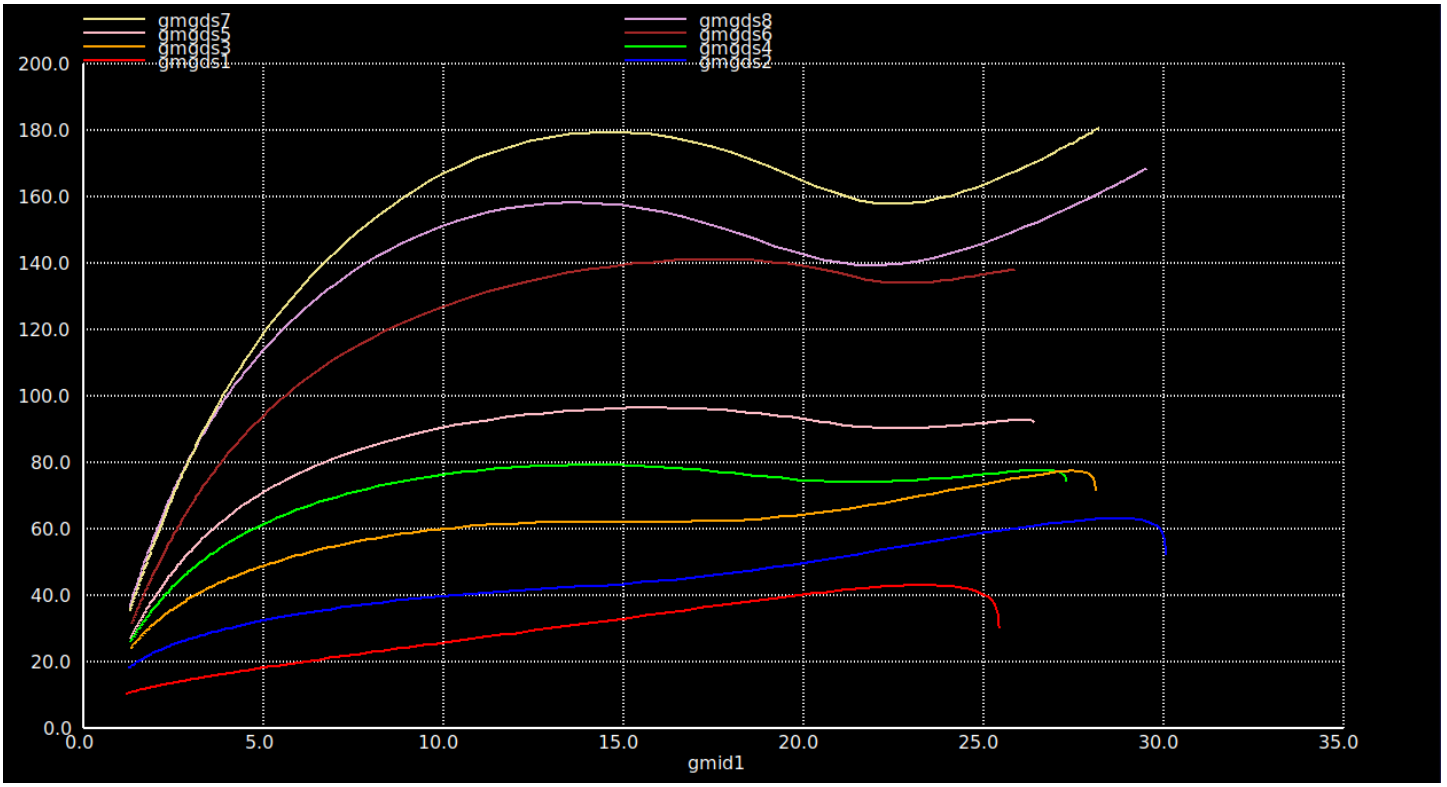
I. g_m/I_D **vs** V_{OV}



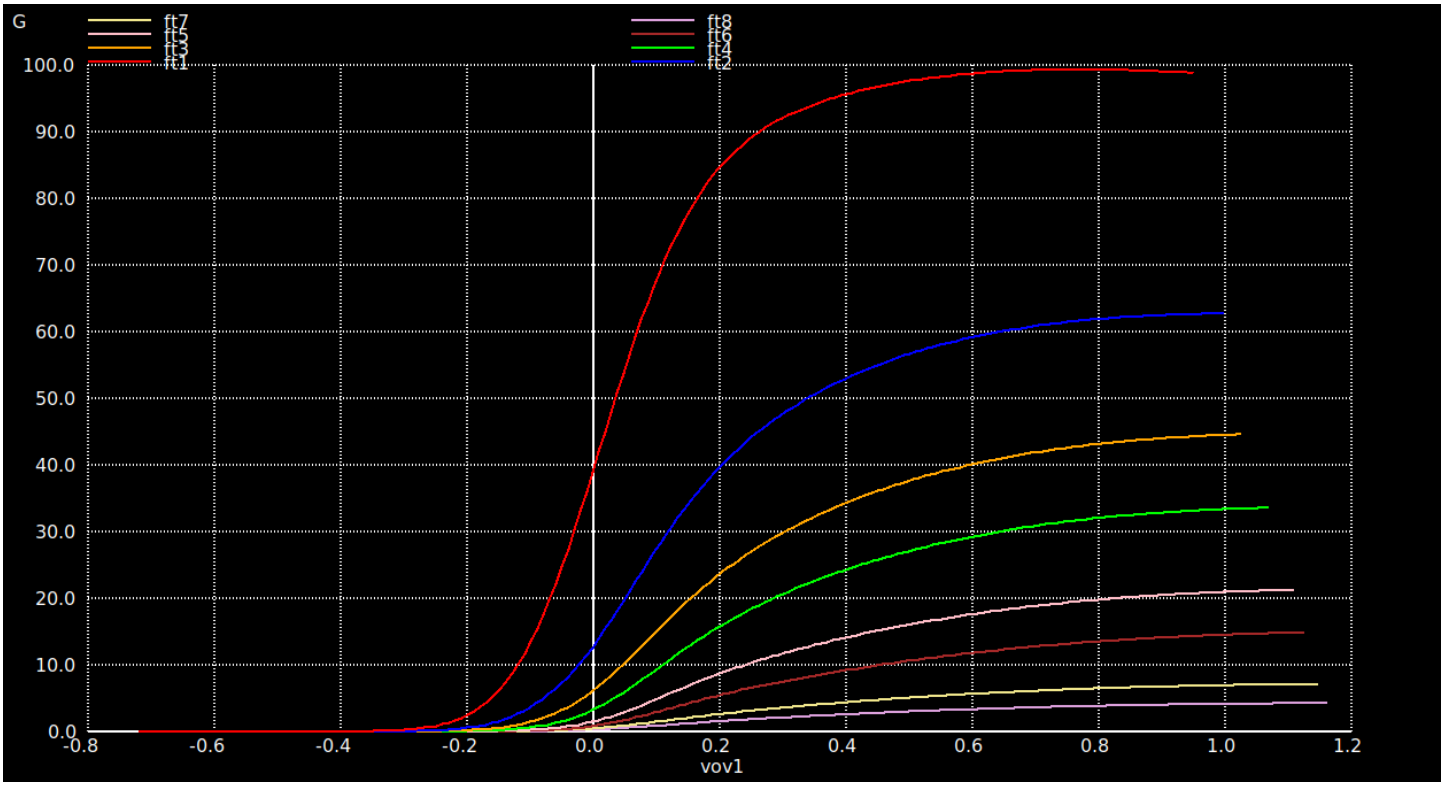
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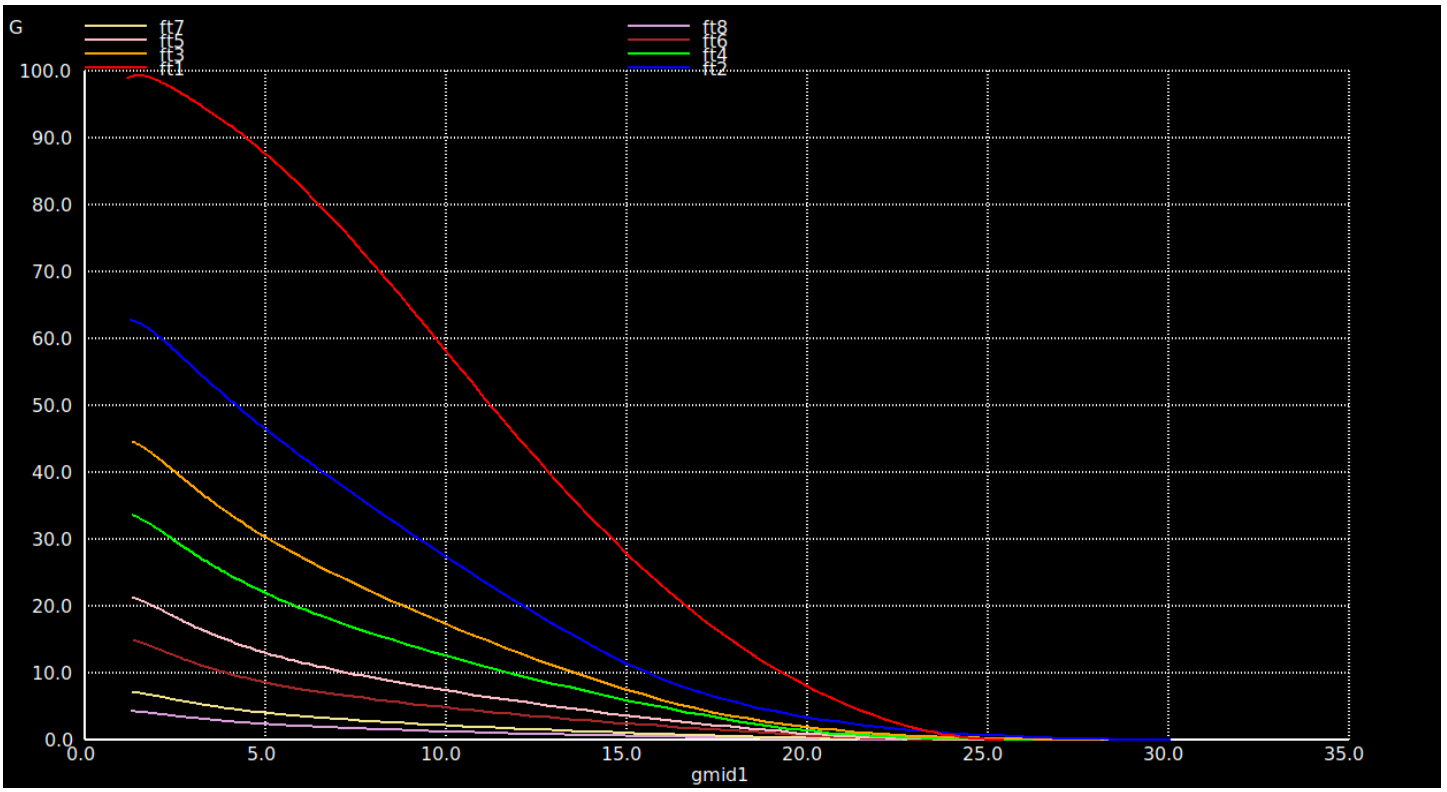
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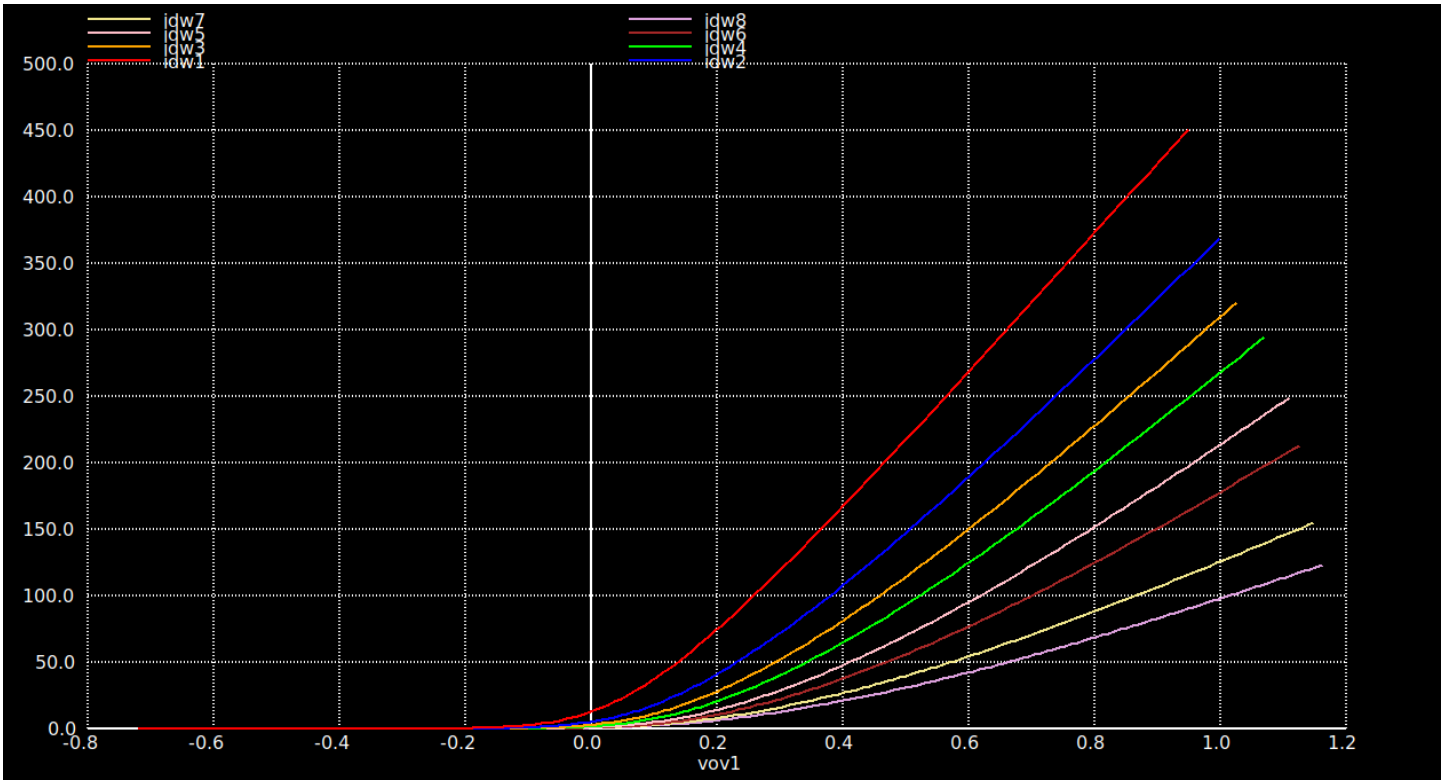
IV. f_T vs V_{OV}



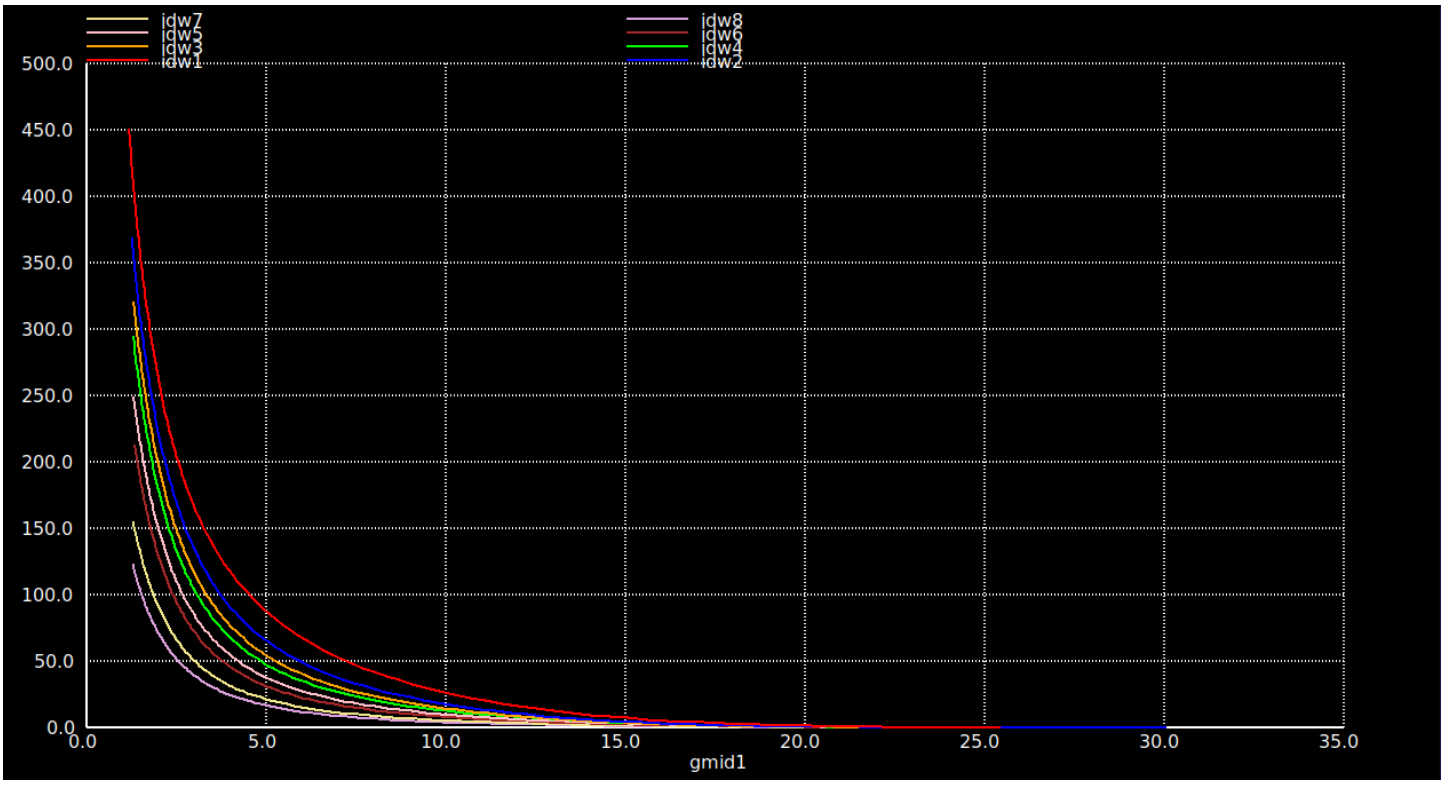
V. f_T vs g_m/I_D



VI. I_D/W vs V_{OV}

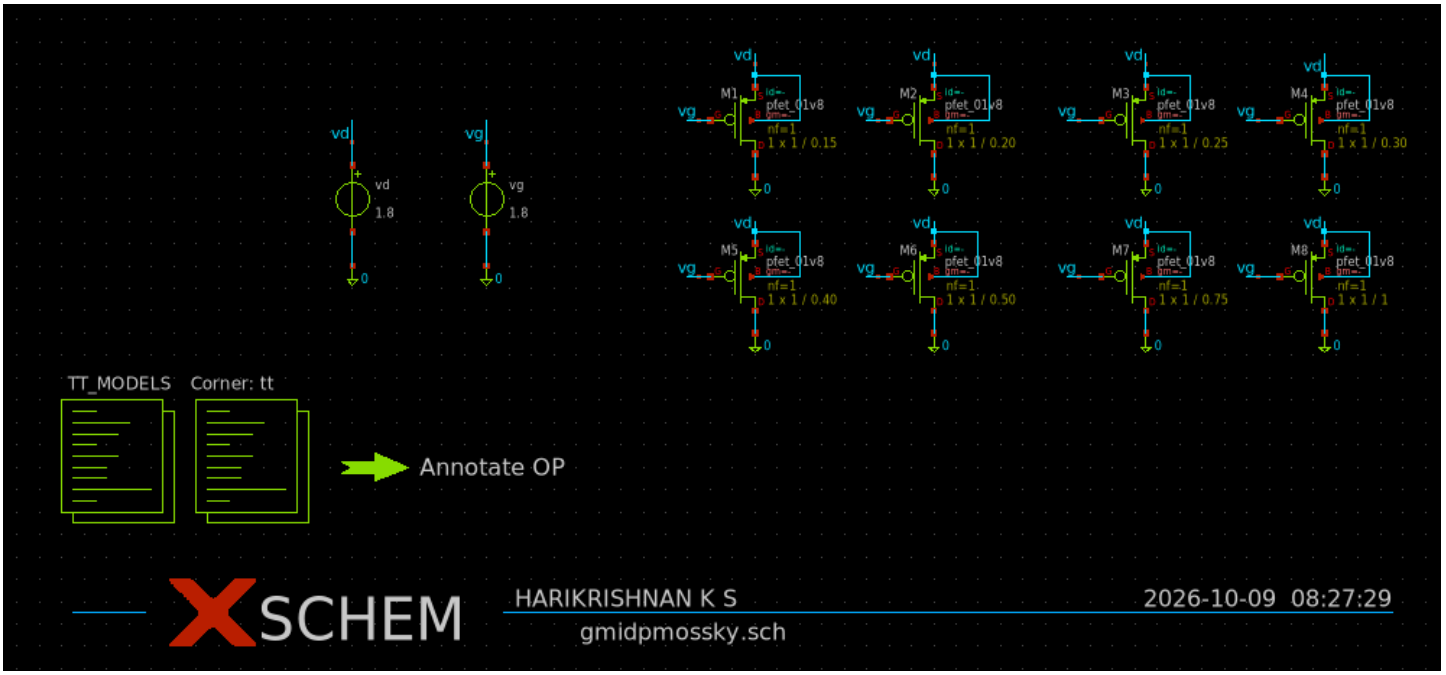


VII. I_D/W vs g_m/I_D



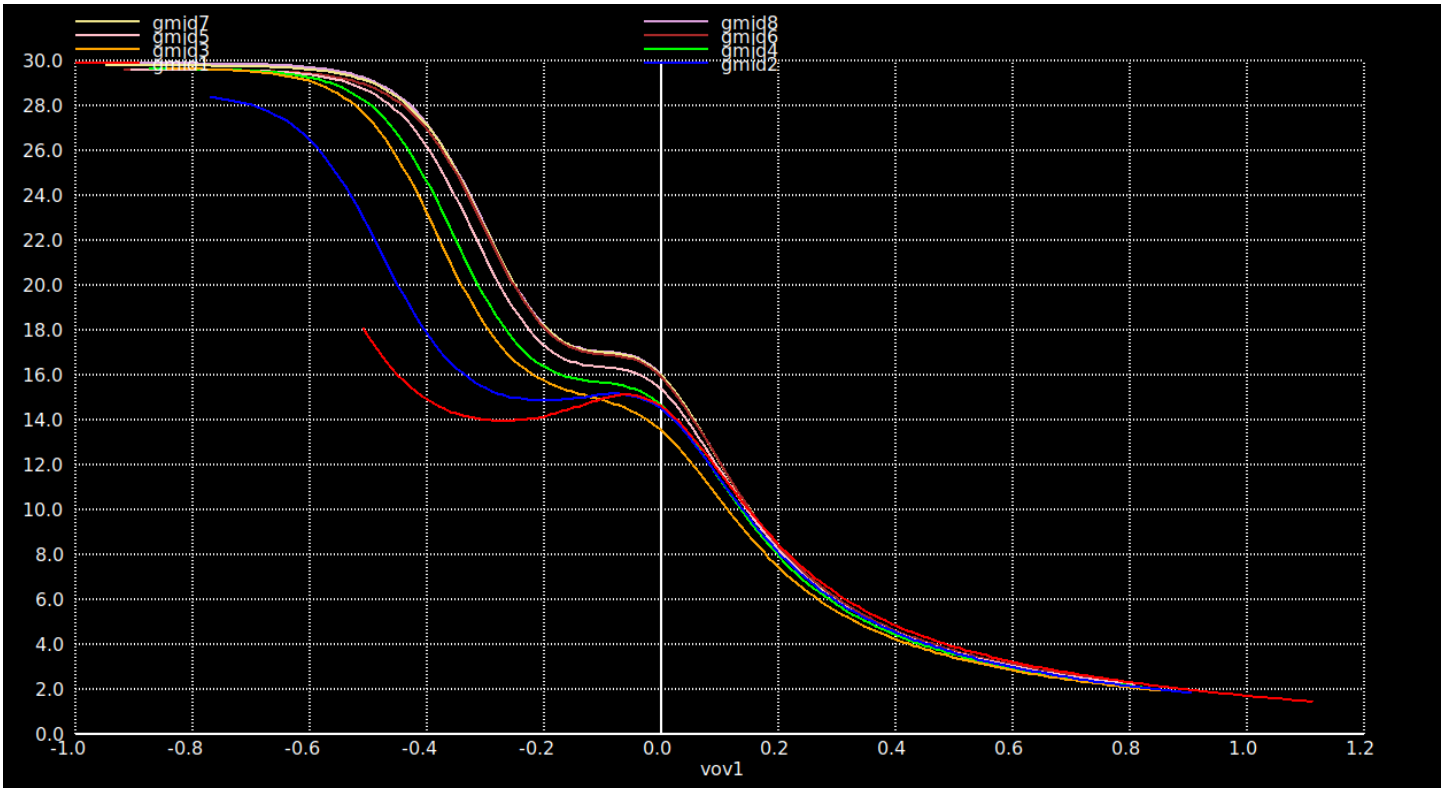
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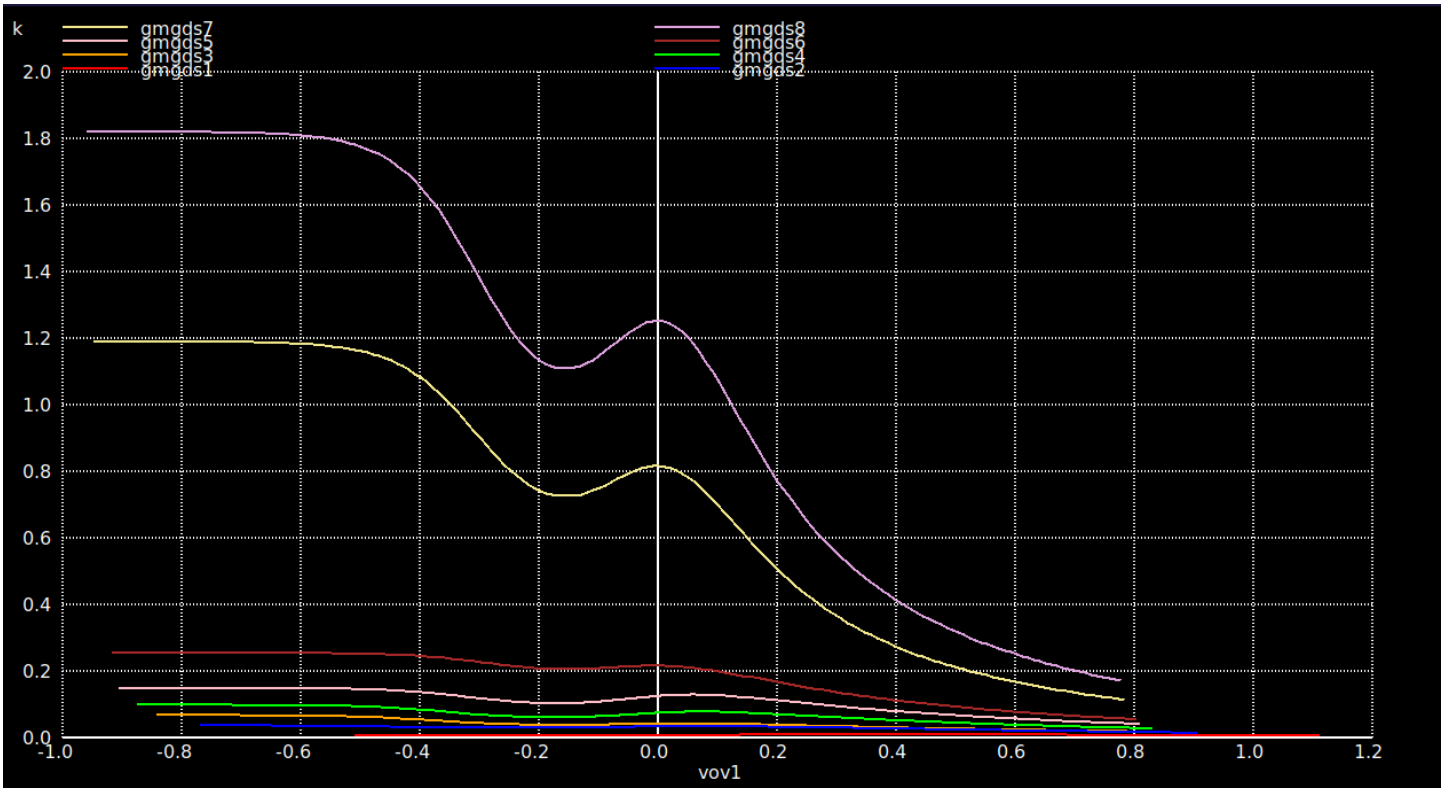


Schematic for Pmos Design charts

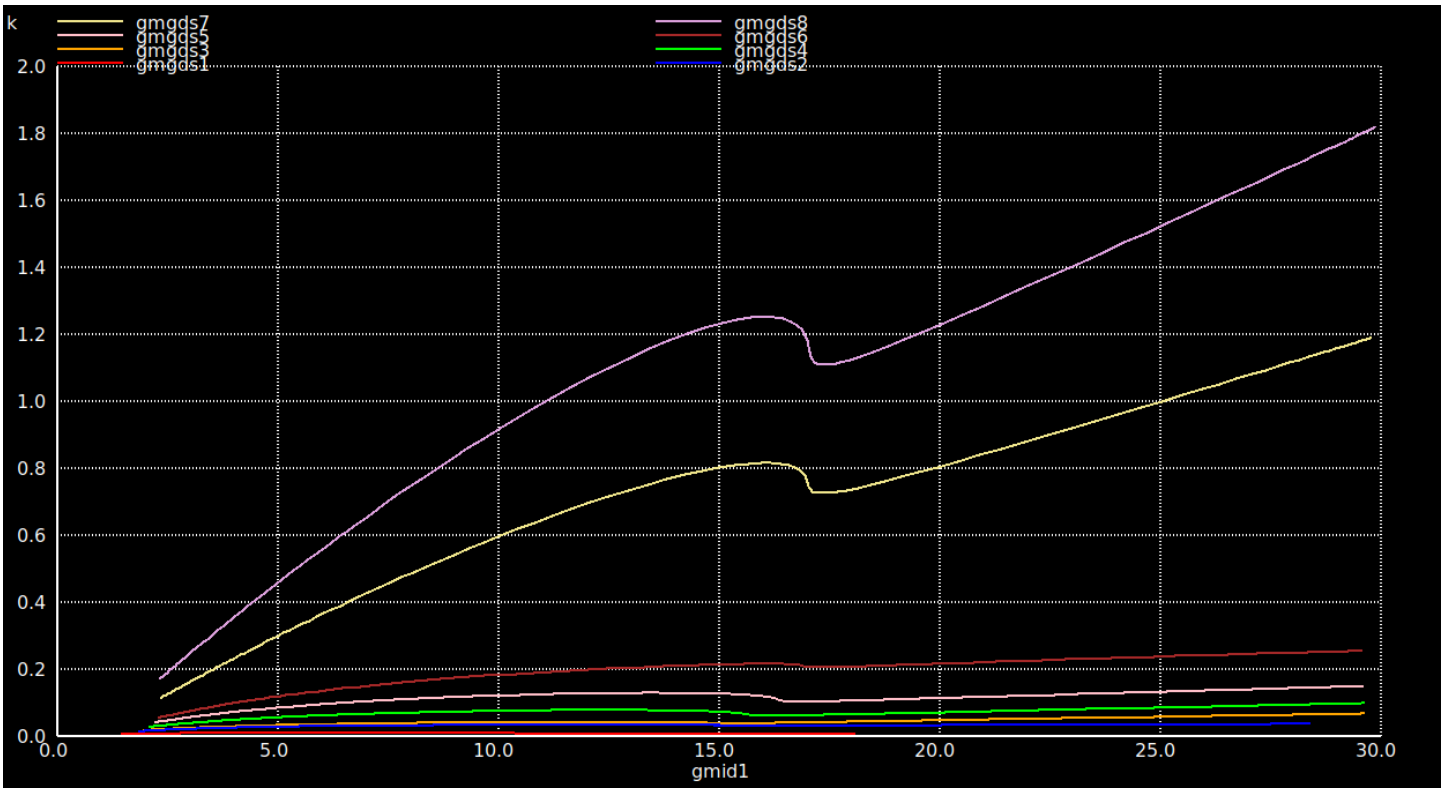
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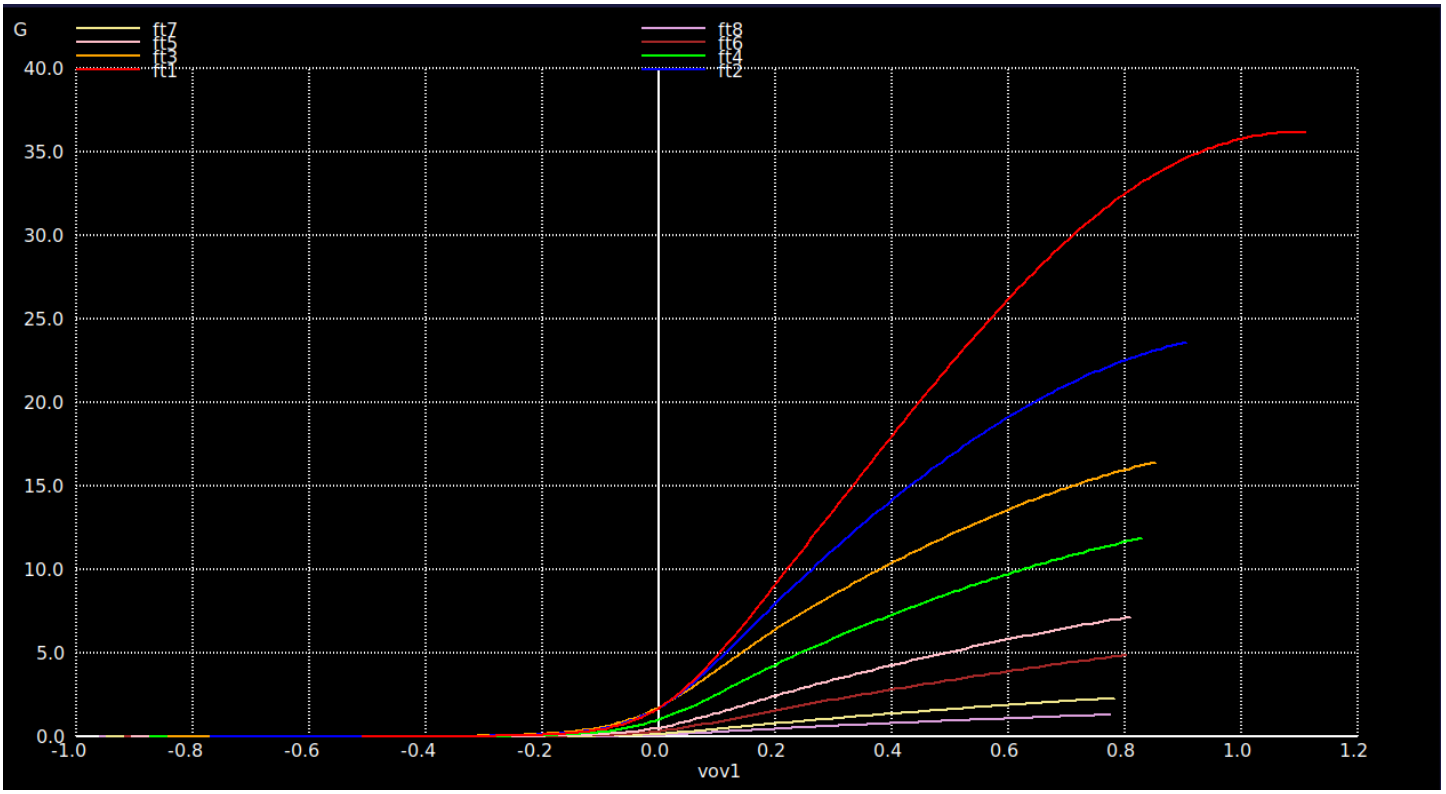
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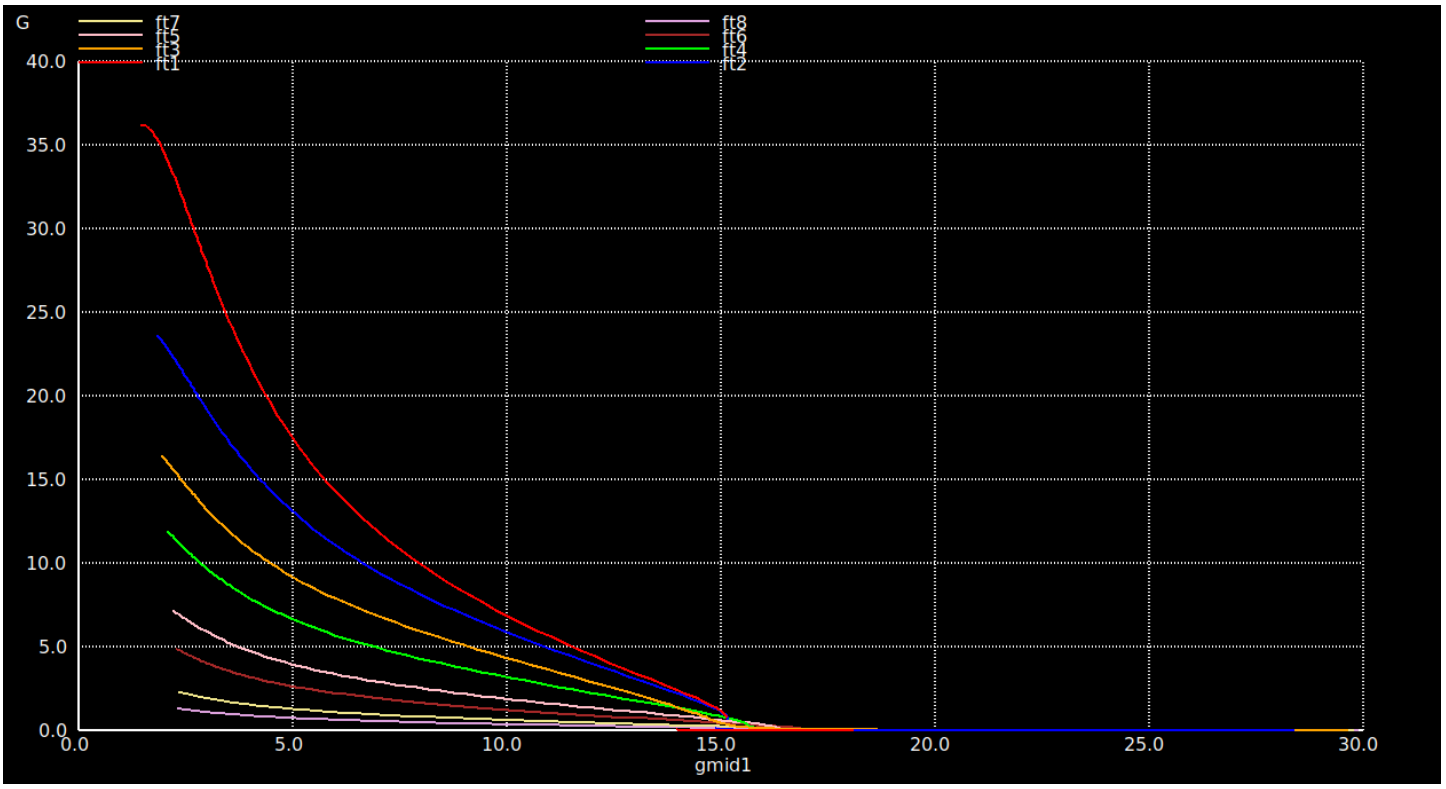
III. g_m/g_{ds} **vs** g_m/I_D



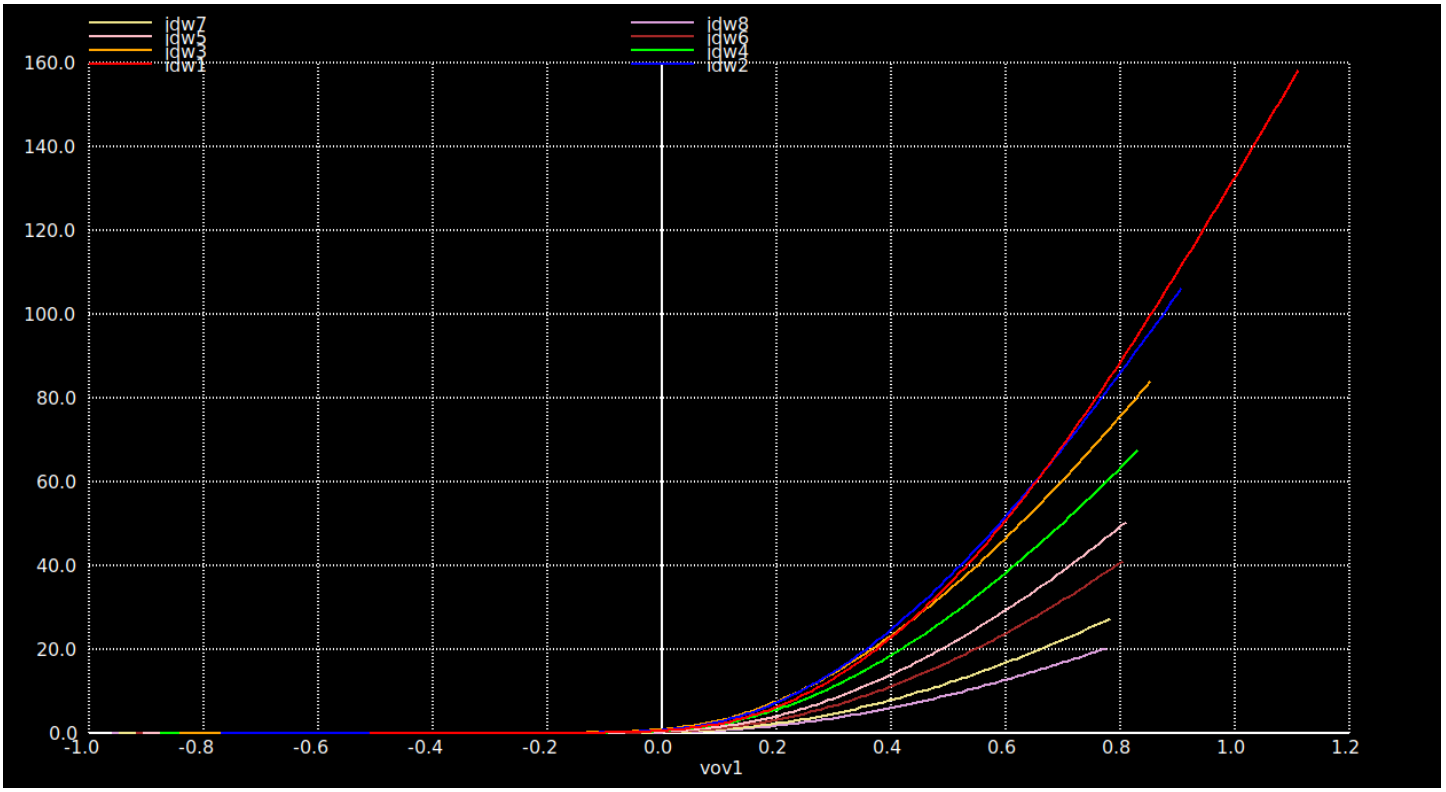
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